# Progress Report — Part 0 Complete

"What Crosses the Cut?" BP-Free Split-Learning Benchmark

Validating the measurement pipeline before the main experiments

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## 1. What this stage was for

Before running the full benchmark, we needed to confirm that the shared testing framework can run different split-learning methods on the same footing and measure all four of our metrics — communication, accuracy, memory, and latency — consistently. This is the "smoke test": a small, deliberately cheap check on a simple dataset (MNIST) whose only job is to prove the machinery works before we commit real compute to the CIFAR-10 and CIFAR-100 experiments. No benchmark conclusions are drawn at this stage.

## 2. What was done

- **Stood up the shared framework** and confirmed it runs end-to-end on a standard method and dataset.
- **Separated the communication measurement into two parts.** The framework originally reported communication as a single combined number. We split it into the two things that matter for our comparison: *cut* communication (the activations and gradients exchanged across the split point) and *weight* communication (the model weights exchanged for federated aggregation), plus their total. This is the central measurement improvement — without it, we could not tell *why* one method communicates more or less than another.
- **Added MNIST** as a test dataset, reusing the exact same model as the main experiments with no changes to the model itself.
- **Added the remaining metrics** — end-to-end latency and peak memory — so every run now records all four metrics to a results file automatically.
- **Ran two different methods to completion** on MNIST and confirmed the measurements behave sensibly and differently across them.

## 3. Results

Both methods trained successfully on MNIST and produced complete, consistent measurements. The table compares a conventional baseline (SplitFedv2) against a backpropagation-free method (CSE-FSL) under identical settings.

| Measurement | SplitFedv2 (baseline) | CSE-FSL (BP-free) |
|---|---:|---:|
| Communication — cut (MB) | 1875.0 | 192.0 |
| Communication — weights (MB) | 20.9 | 85.1 |
| Communication — total (MB) | 1895.9 | 277.1 |
| Cut vs. weights share | ∼99 : 1 | ∼69 : 31 |
| Test accuracy | 98.8% | 98.4% |
| Latency (s) | 538.5 | 350.6 |
| Peak memory (MB) | 1060.8 | 1113.4 |

**What the numbers show.** The two methods behave exactly as their designs predict, which tells us the measurements are capturing something real:

- CSE-FSL used about **7× less total communication** than the baseline — consistent with the fact that it exchanges full gradients only periodically rather than every step.

- The **cut-vs-weights breakdown differs sharply** between the two methods: the baseline's communication is almost entirely cut traffic, while CSE-FSL shifts a much larger share to weight traffic (from the extra model it aggregates each round). A single combined number would have hidden this difference entirely.
- **Latency dropped** in step with the reduced work, and both methods reached comparable accuracy — expected on a simple dataset.

This confirms the measurement layer discriminates between methods correctly, rather than producing a constant offset.

## 4. What this proves

1. **The shared framework holds up.** One common testing setup now runs multiple methods across two datasets with no special-casing — the fair, common measurement basis the benchmark's contribution depends on.
2. **All four metrics are measured uniformly** for every method and saved automatically to a results file, ready for the ranking and plotting stage.
3. **The communication split is meaningful, not cosmetic** — it reveals *why* methods differ, which the paper's analysis needs, not just *that* they differ.

## 5. Notes and next steps

- We found and documented a pre-existing bug in one of the framework's default model configurations; it should be fixed (or that model avoided) before the main runs.
- These runs were small, CPU-only validation runs on a development machine — the numbers confirm the measurement works, but are not representative benchmark results.
- **Next:** scale to CIFAR-10 with short runs to confirm the pipeline holds on the real dataset, then begin populating the benchmark table (full training, multiple seeds, the data-heterogeneity settings).

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*Bottom line:* the measurement pipeline works. All four metrics — with communication correctly separated into cut versus weights — log uniformly across a baseline and a backpropagation-free method and distinguish them in a way that matches how the methods actually behave. Part 0 is complete; we are ready to scale to the real datasets.